

Pose estimation model selection for on-device mobility assessment

Case study, Task 2. Which pose estimator to run on the user's phone for an overhead-squat capture, given the constraints: video stays on the device, minimal compute, and landmarks good enough for joint measurement. Test material: **Example1.mov** (720x1280, 30 fps, 54.9 s, one subject, three squat holds: frontal, side facing left, side facing right). Every candidate ran on the same 550 frames (every 3rd frame).

1. Summary

- **Recommendation:** RTMW-m through **rtmlib** + ONNX Runtime. 133 keypoints (body, 6 foot points, 21 per hand, face), ~36 ms per frame on a desktop CPU, 25 ms amortised once the person box is tracked instead of re-detected. Apache-2.0, convolutional backbone, ships as ONNX.
- Foot keypoints (heel, big toe, small toe) are a hard requirement for this use case: the subject's heels leave the floor in every squat, and a model without foot points cannot see that. This rules out Apple Vision and MoveNet.
- Lightweight monocular 3D was not reliable enough to build on. Each small model's 2D landmarks were more consistent than its own 3D output, so the pipeline is designed around 2D landmarks in the side views.
- Heavy models (SAM3D, Sapiens-1B) are GPU-only, 1–2 GB of weights, and under restrictive licences. They are kept as offline references for auditing the on-device model, never shipped.
- The joint-level accuracy evaluation (angles against the reference models and the research bands) is documented separately in the notebook and findings JSON.

2. Requirements the model must meet

- **Runs on the phone.** No video leaves the device, so no server-side inference. Target: tens of ms per frame on a mobile CPU/NPU; a few tens of MB of weights.
- **Foot landmarks.** Heel and toe points are needed for ankle measurement and heel-lift detection. Hip, knee, ankle alone cannot distinguish a lifted heel from a flat foot.
- **Per-landmark confidence.** In a side view the far-side limbs are hidden. The model must say so rather than guess.
- **Fingers** are useful, not required: hand points allow wrist/grip measures and reduce elbow-related ambiguity in upper-limb measurement.
- **Permissive licence** for commercial on-device use.

3. Candidates tested

Pose-step time is the median per frame for the keypoint model alone. The detector column is the person-box detector each pipeline used; on a phone it runs once and the box is then tracked from the previous frame. CPU = desktop CPU (ONNX Runtime / MediaPipe). GPU = RTX 4090.

Model	Kpts	Feet	Hands	Output	Pose ms	Detector ms	Weights / licence
SAM3D (Fast SAM 3D Body, DINOv3 ViT-H+)	70	heel + 2 toes	21/hand	3D metric (mesh-derived)	~45 GPU	YOLO11m	953M params, 2.1 GB; Meta SAM licence, gated
Sapiens-1B (Goliath-308)	308	yes	21/hand	2D heatmaps	82 GPU fp16	n/a (reused boxes)	4.7 GB; CC-BY-NC-4.0
RTMW3D-x	133	yes	21/hand	2D + depth	150 CPU	170 (YOLOX-m)	large only; 3rd-party ONNX host
RTMW-I 384x288	133	yes	21/hand	2D	101 CPU	170 (YOLOX-m)	Apache-2.0
RTMW-m 256x192	133	yes	21/hand	2D	36 CPU	28 (YOLOX-tiny)	Apache-2.0
RTMPose-x (Halpe-26)	26	yes	no	2D	97 CPU	449 (YOLOX-x)	Apache-2.0
RTMPose-m (Halpe-26)	26	yes	no	2D	18 CPU	167 (YOLOX-m)	Apache-2.0
RTMPose-s (Halpe-26)	26	yes	no	2D	6.6 CPU	25 (YOLOX-tiny)	Apache-2.0
BlazePose heavy	33	heel + toe	3 pts/hand	2D + world 3D	151 CPU	built in	MediaPipe, Apache-2.0
BlazePose full	33	heel + toe	3 pts/hand	2D + world 3D	49 CPU	built in	Apache-2.0
BlazePose lite	33	heel + toe	3 pts/hand	2D + world 3D	36 CPU	built in	Apache-2.0
Apple Vision body pose	17–19	no	no	2D / 3D	NPU	built in	free; iOS only
MoveNet	17	no	no	2D	fast	built in	Apache-2.0

Apple Vision and MoveNet were excluded on the foot-landmark requirement and not run. Every model that was run detected the subject in all 550 frames.

4. Observations from the runs

Detection and tracking

- No candidate lost the subject: 550/550 frames for all eleven models run, through walking, turning and the three holds.
- The person detector dominates CPU cost in the RTM pipelines (25–449 ms vs 7–150 ms for pose). Running it every 30 processed frames and tracking the box from the previous keypoints brought RTMW-m's amortised cost to 25 ms per frame with 19 detector calls over the whole video, and no lost frames.

Occlusion and confidence

- Side views hide the far-side arm and leg. BlazePose and Sapiens lowered confidence on those landmarks clearly (BlazePose: hidden wrist 0.09 vs 0.93 on the visible side; Sapiens: 0.4–0.5 vs 0.9–1.0).
- RTMPose-m and RTMW did not always lower confidence on a collapsed far-side leg; in the comparison video RTMPose-m placed a hidden hip on the knee while still reporting it as confident.
- SAM3D provides no per-joint confidence; occluded limbs are rendered as plausible guesses with nothing to flag them.
- Design consequence: the measurement pipeline uses the near-side limbs only in side views, and treats the confidence gate as a second filter rather than the sole safeguard.

2D versus 3D output

- BlazePose's world (3D) landmarks were visibly wrong at the bottom of the squat and unusable at the feet, while its 2D landmarks were reasonable. Its lite model's foot points were unreliable even in 2D.
- RTMW3D-x produced the most stable lightweight 3D, but depth remained its weak axis, and its output mixes crop pixels with metres, requiring a scale assumption before it is a consistent skeleton.
- Across the RTM family and Sapiens, 2D landmark placement on the visible side was consistent from the 6.6 ms model up to the 1B-parameter model; model size bought little on this footage.
- SAM3D's 3D output is the most trustworthy, including the feet (its heel and toe points come from a body mesh), but it requires a datacentre-class GPU and a 2.1 GB checkpoint.

Video conditions that stressed the models

- Phone on the floor, pitched upward: strong perspective, so the image vertical is not the true vertical.
- Backlit doorway between the legs; loose shorts hiding the hip; coats and a shelf behind the hands in the side views.
- About 70% of the footage is walking and turning, so hold detection (from the keypoints) is where the compute saving is: the pose model only needs to run densely inside ~15 s of holds.

5. Why RTMW-m

- **Meets every requirement:** feet and hands, per-landmark scores, 2D output that was as consistent as the heavy models', Apache-2.0, ONNX, convolutional backbone (CoreML/TFLite-friendly; conversion not yet done).
- **Cost:** 36 ms pose + tiny detector on a desktop CPU; 25 ms amortised with box tracking. RTMW-I, RTMPose-x and RTMW3D-x cost 3–4x more for no observable gain on this footage.
- **Versus RTMPose-m** (18 ms, 26 keypoints, same accuracy class): the better choice if hands are not needed. RTMW-m is preferred because the hand points cost 18 ms and remove a future re-integration.
- **Versus BlazePose:** simplest to integrate (MediaPipe) and has per-landmark visibility, but its 3D is unreliable and its 2D landmark placement on the leg was the outlier among the models tested.
- **Versus SAM3D / Sapiens:** better landmarks, but GPU-only, gigabyte checkpoints, and licences (Meta SAM licence, CC-BY-NC) that block on-device shipping. They remain the offline reference and teacher models: run on consented internal footage only, never on user video.

6. Deployment plan and open items

- Convert RTMW-m and YOLOX-tiny to CoreML (or TFLite) and measure on a target phone; the timings above are desktop-CPU.
- Detector once per capture plus keypoint-based box tracking; re-detect on tracking loss.
- Run the pose model at reduced frame rate outside the holds; densely inside them.
- Keep SAM3D (3D) and Sapiens (2D) as offline audit models for internal captures, to check the shipped model over time.
- Confidence handling for the RTM family needs a near-side rule, since its scores do not always fall on occluded limbs.
- Single video, single subject: repeat the detection, confidence and timing checks on a wider set of captures before committing.